Generative AI and scenario generation in insurance

What does “extreme” mean?

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Motivation

Motivation: generation of extreme scenarios for insurance

Risk from catastrophes (both natural and otherwise) is driven by events which are both rare and high-dimensional, yet in most cases, the record only contains a tiny number of representative events.

- Losses are typically **heavy-tailed** where only a few events dominate aggregate risk
- **Dependence structure** matters. Losses across coordinates are often more important than marginal extremes alone
- E.g.,: spatially distributed hazards such as **drought** and **hail**

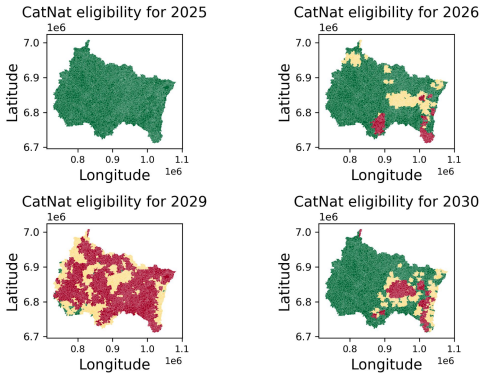


Figure 1: Generated eligibility for catastrophe recognition due to soil subsidence from drought under RCP 8.5 in the Grand Est region of France. Heranval et al. (2025)

We need generative models which are reliable in the joint tail with scarce observations

Extreme value theory revision: the univariate case

The Generalized Pareto Distribution (GPD)

Definition (Generalized Pareto Distribution (GPD))

A random variable Y follows a Generalized Pareto Distribution with shape $\xi \in \mathbb{R}$ and scale $\sigma > 0$ if

$$\Pr(Y \leq y) = H_{\xi,\sigma}(y) = \begin{cases} 1 - \left(1 + \xi \frac{y}{\sigma}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\frac{y}{\sigma}\right), & \xi = 0, \end{cases}$$

for $y \geq 0$ such that $1 + \xi y/\sigma > 0$.

The Generalized Pareto Distribution (GPD)

The shape parameter ξ defines three key regimes:

	Weibull $\xi < 0$	Gumbel $\xi = 0$	Fréchet $\xi > 0$
Tail decay	finite endpoint	exponential	polynomial (heavy)
Moments	all finite	all finite	finite only up to $1/\xi$

The peaks-over-threshold approach to modelling extremes

Theorem (Pickands–Balkema–de Haan)

For a random variable $X \in \mathbb{R}$, for a sufficiently high threshold u, the distribution of exceedances is well approximated by a Generalized Pareto Distribution:

$$\Pr(X - u \leq y \mid X > u) \approx H_{\xi,\sigma}(y), \qquad y \geq 0.$$

Instead of examining X directly, we focus on the **conditional excess** $Y = X - u$, given $X > u$.

Modelling extremes: from the CLT to EVT

	Central Limit Theorem	Extreme Value Theory	
		Block maxima	Peaks over threshold
Quantity	Sums of i.i.d. random variables	Block maxima	Threshold exceedances
Part of the distribution	Bulk / centre	Tail	Tail
Limiting family	Gaussian	GEV	GPD
Modelling choice	Sample size	Block size	Threshold u

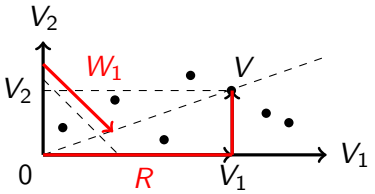
In this project we examine the peaks-over-threshold approach and choosing u.

Extreme value theory revision: the multivariate setting

Multivariate extremes: polar decomposition

Let $X = (X_1, \ldots, X_d) \in \mathbb{R}_+^d$ a random vector.
We define $V_i = (1 - \hat{F}_i(X_i))^{-1}$ the unit-Pareto margins with $\hat{F}_i(X_i)$ the empirical c.d.f of X_i.
We apply the polar decomposition:

$$R = \|V\|_1 = \sum_{i=1}^{d} V_n, \qquad W = \frac{V}{R}.$$



- Radius R: magnitude / severity of an event
- Angle W: direction / dependence across coordinates

Goal: estimation of both the radius R and the angle W

Multivariate extremes: asymptotics

Define the radial log excess ratio

$$E = \log(R/u).$$

Under the assumption of **multivariate regular variation** we have the asymptotic property[1]:

$$\mathcal{L}(E, W \mid R > u) \to \text{Exp}(1) \otimes H, \quad u \to \infty.$$

If $R > u$ for large enough u, $R \perp W$. We can thus estimate R and W separately.

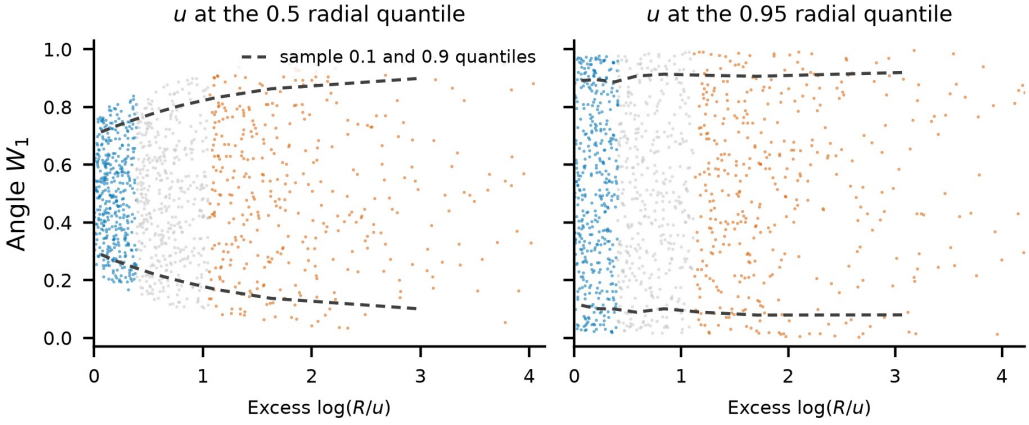
- R estimated by well-established univariate techniques
- **W** estimated with flexible generative methods e.g., Wasserstein GANs

Problem: How high should the threshold u be set?

[1] Resnick, 2007

Multivariate extremes: asymptotics

Take a sample of N realisations.
Choose a threshold u and check the independence.



Modelling: the extremal threshold u

Choosing the threshold

The GPD approximation is only asymptotic in u.
Choosing u is the central practical difficulty of the method.

u too small

- retained values are not really extreme
- GPD approximation is biased

u too large

- too few exceedances left
- estimator variance explodes

No free lunch: u trades bias against variance.

Synthetic data

Why do we need synthetic data?

On real data, that ground truth does not exist. We do not know the true law of X, so we cannot tell whether:

- a threshold-selection procedure has actually found a good threshold,
- a generative model has correctly reproduced the bulk and the tail

Synthetic data: by constructing X ourselves, we control the full joint law of the data, so any gap we observe is attributable to the performance of the method being tested, not to an unknown truth.
Challenge: how to construct X such that it has similar properties to the real-world data driving our motivation (heavy-tailed, complex dependence structure and multimodality)?

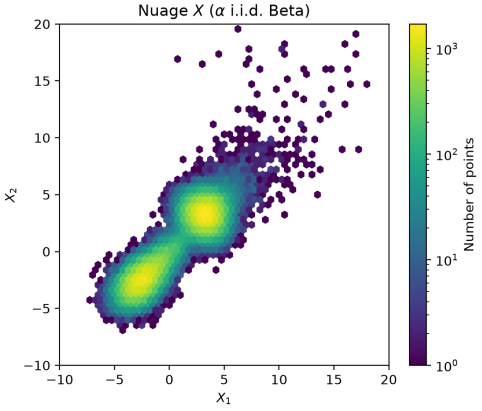
Synthetic dataset 1

We construct X as the (random) mixture of an **extremal** and **bulk** distribution:

$$X = \alpha(Z + u) + (1 - \alpha)G$$

- Z: MGPD $\rightarrow$ **extremes** (heavy tail)
- G: bimodal Gaussian mixture $\rightarrow$ **body**
- α: Beta distributed$\rightarrow$ **mixing weight**

When α is small $\Rightarrow$ G dominates (bimodal)
When α is large $\Rightarrow$ Z dominates (heavy tail)

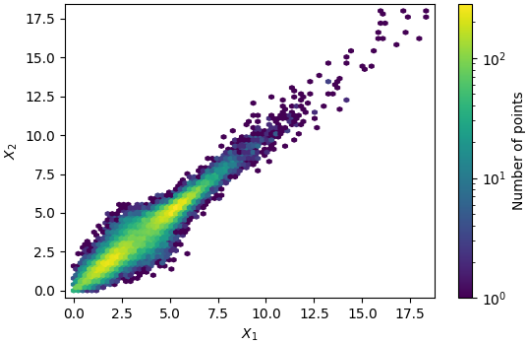


Synthetic dataset 2: univariate laws + copula

Sklar's Theorem: The joint law F admits an unique decomposition between univariate laws F_i and a copula C: $F(x) = C(F_1(x_1), \dots, F_d(x_d))$.

- Copula: sample N realisations of $U = (U_1, \dots, U_d)$ with c.d.f C such that $U_i \sim \mathcal{U}([0,1])$
- Marginal laws: F_i for $i \in \{1, \dots, d\}$

$$(X_1^n, \dots, X_d^n) = (F_i^{-1}(U_1^n), \dots, F_i^{-1}(U_d^n))$$

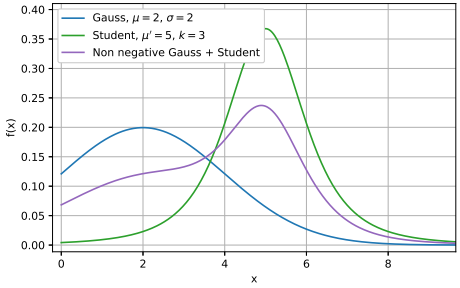


- Total control over the wanted synthetic data (marginal and dependence)
- Synthetic data with no clear threshold (bulk and tail).

Synthetic dataset 2: univariate laws + copula

Univariate probability density function:
Gauss $Z \sim \mathcal{N}(\mu, \sigma)$ and Student $S \sim S_t(\mu', k)$ p.d.f.

$$\mathbf{f(x)} = p(f_Z(x) + f_S(x))1_{\{x \geq 0\}}$$

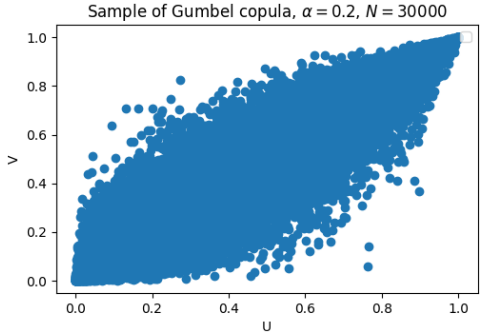


Complex bulk, heavy tail $\xi = \frac{1}{k}$

Dependence structure: Gumbel copula $\alpha \in (0, 1]$

$$(\mathbf{U_i})_\mathbf{i} = \left(\exp\left(-\left[\frac{E_i}{S}\right]^\alpha\right)\right)_i,\ U_i \sim \mathcal{U}([0,1]).$$

with $E, E_1, \ldots, E_d \sim \mathcal{E}(1)$
and S sampled from positive stable law.



Extreme Value copula

Returning to the problem of choosing the extremal threshold u

Reminder

The conditionally independent radial–angular model is asymptotic. For 1-Pareto margins, we have:

$$\mathcal{L}(E, W \mid R > u) \Longrightarrow \text{Exp}(1) \otimes H, \qquad u \to \infty.$$

However in practice, we need to must *finite* threshold u to model the extremes.

u too low

- more exceedance observations
- radial–angular dependence may remain
- modelling based on radial-angular independence may be inappropriate

u too high

- fewer exceedance observations
- less information for estimation and generation
- stronger case to utilise the asymptotic independence property

Can examining the radial–angular depedendence itself tell us how to choose u?

Independence testing via radial-angular permutations

A finite-threshold condition

Given a fixed threshold u, consider the null hypothesis

$$\boxed{\mathcal{H}_0(u): \qquad E \perp W \mid R > u}$$

equivalently

$$\mathcal{L}(E, W \mid R > u) = \mathcal{L}(E \mid R > u) \otimes H_u.$$

Conditional independence of the radius and angle gives a testable sufficient condition for a stable angular measure.

Testing independence via permutations

Consider m observations exceeding u. Decompose these into radial execess and angle pairs

$$(E_1, W_1), \ldots, (E_m, W_m).$$

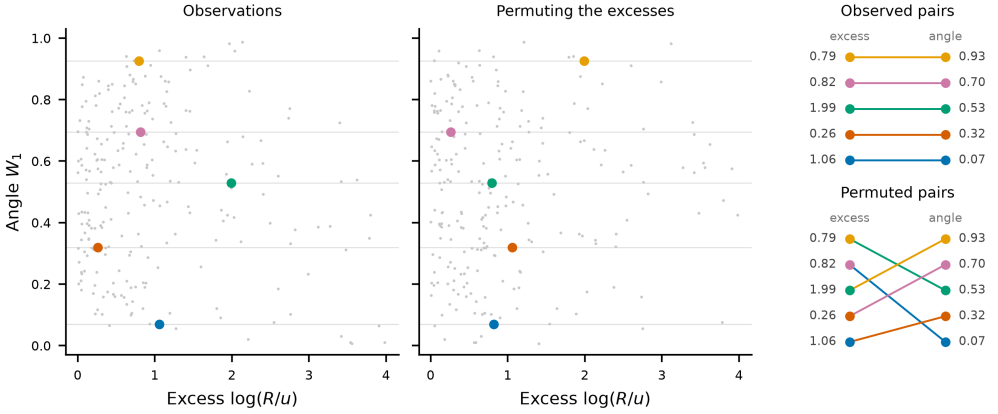
Proposition (distributional invariance under permutation)

Under $\mathcal{H}_0(u)$ (conditional radial-angular independence), the observed excesses $E_{\sigma(i)}$ can be re-paired with angles W_i without changing the joint law of the sample. For any permutation σ:

$$\boxed{(E_{\sigma(i)}, W_i)_{i=1}^m \overset{d}{=} (E_i, W_i)_{i=1}^m}$$

- Permutation preserves the marginal distributions of E and W and, under independence, the joint law of (E, W) whilst removing the original pairing
- **Repeated permutation generates samples under the null of radial-angular independence**

Permutation illustration



Under the null of independence, the observed pairing has the same distribution as any randomly re-paired sample.

The permutation independence test

Choose a fixed dependence statistic T which is larger when stronger dependence is detected.
Evaluate T on the original exceedance observations,

$$T_{\mathrm{obs}} = T\big((E_i, W_i)_{i=1}^{m}\big).$$

Also evaluate T on B independent random permutations,

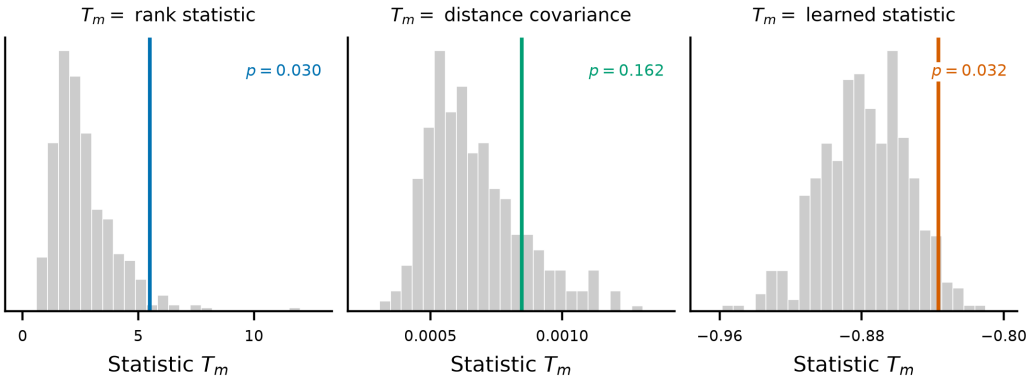
$$T_b = T\big((E_{\sigma_b(i)}, W_i)_{i=1}^{m}\big), \qquad b = 1, \ldots, B.$$

Definition (permutation p-value)

$$\widehat{p} = \frac{1 + \#\{b : T_b \geq T_{\mathrm{obs}}\}}{B + 1}.$$

T_{obs} is compared against the distribution of dependence statistics T_b after the radius–angle pairing is broken, with the permutation p-value defined empirically.

The test statistic can be chosen freely



Finite-sample null calibration

Proposition (Finite-sample type-I error calibration)

Under $\mathcal{H}_0(u)$,

$$\mathbb{P}(\widehat{p} \leq \alpha) \leq \frac{\lfloor \alpha(B+1) \rfloor}{B+1} \leq \alpha.$$

the permutation test has a finite-sample type-I error rate of at most α.

- Contrasting from competing approaches, no asymptotic approximation of the null distribution is used
- The result holds for any number of exceedances m and permutations B[2]
- It does not rely on any specific dependence statistic

Permutation determines calibration; the choice of statistic determines power.

[2]Though the empirical null distribution becomes coarse and of little use with small B or m).

Multiple testing for threshold selection

Multiple testing for threshold selection

Given a specified set of threshold candidates

$$u_1 < u_2 < \cdots < u_G$$

compute their corresponding permutation test p-values

$$p_1, \ldots, p_G.$$

Testing across a range of thresholds is a multiple-testing problem. We therefore need a Bonferroni $p - value$ correction and thus reject at a threshold u_g only when

$$p_g \leq \frac{\alpha}{G}.$$

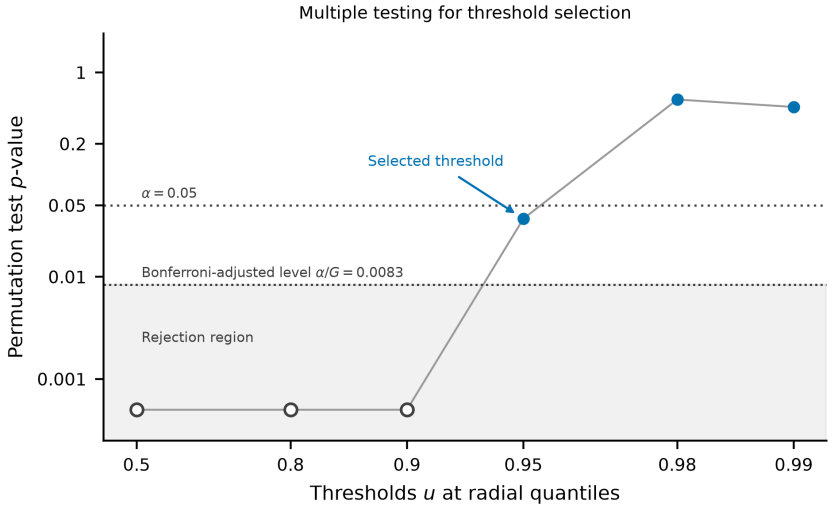
Then the lowest threshold which fails to be rejected is selected:

$$\boxed{\widehat{u} = \min\left\{u_g : p_g > \frac{\alpha}{G}\right\}.}$$

The Bonferroni correction gives the guarantee

$$\mathbb{P}(\text{rejecting at least one candidate where the null is actually true}) \leq \alpha.$$

Threshold selection in practice



The threshold moves into the tail whilst dependence is detected until the first candidate above the adjusted rejection region.

What does the selected threshold represent?

Multivariate regular variation induced independence is an asymptotic condition,

$$\boxed{\text{failure to reject independence } \neq \text{ proof of full independence}}$$

The threshold selected can be interpreted as

The lowest candidate threshold for which the exceedances $(E_i, W_i)_{i=1}^m$ and the chosen statistic T_m do not detect significant radial–angular dependence at the chosen testing level α.

- False rejection of a threshold where independence already holds (type-I error) is directly controlled
- Failure to detect residual dependence is a question of power
  - Power depends on the number of exceedances m, dimension d and the statistic T_m

Some assumptions

The presented finite-sample permutation argument is exact but does have some natural assumptions.

- The observations are i.i.d.
- Marginal standardisation is treated as known and fixed
  - Fitting marginal transformations from observations also used in the independence test can introduce contaminating dependence
  - Marginal transformations estimated from independent data are okay!

Aside: a learned dependence statistic

One attractively general construction of an admissible test statistic is a *learned* scoring function. First split the exceedances into train and test splits, then:

1. Fit a score $S(E, W)$ on the training set aiming to distinguish observations from re-paired samples
2. Fix the learned S;
3. evaluate

$$T = \frac{1}{m_{\text{test}}} \sum_{i \in \text{test}} S(E_i, W_i)$$

   on the test sample and permutations.

Conditional on the training data, S is fixed, so the previous finite-sample permutation procedure applies.

The details of this statistic can be flexibly designed to maximise power.

An alternate approach: avoiding a fixed threshold altogether

Avoiding a fixed threshold altogether

So far, we have treated the threshold as a nuisance parameter to be *chosen*:

$$\text{choose } u \quad \longrightarrow \quad \text{fit a model on } R > u.$$

However we can consider an alternate direction:

Can we learn a smooth transition between the bulk and extremes instead of choosing exactly where it begins?

Example of such a construction:

$$\mathcal{G}(r) = \big(1 - \alpha(r)\big)\mathcal{G}_{\text{bulk}} + \alpha(r)\mathcal{G}_{\text{extreme}}, \qquad \alpha(r) \in [0, 1].$$

With $\alpha(r)$ a smooth, learnable transition toward extremeness.

- smaller radii inform the bulk model
- larger radii inform extremal model
- the transition is learnt rather than fixed at a single threshold

Next steps and takeaways

Next steps

Threshold selection

- Evaluate the power of the permutation test across:
  - Test statistics T_m
  - Sample size m
  - Dimension d
  - Joint laws $\mathcal{L}(R, W)$
- Compare selected thresholds with known angular stabilisation using synthetic data
- Study the procedure's interaction with estimated margins, dependent observations and real-world data

Avoiding a hard threshold

- Explore potential smooth bulk–extreme transition mechanisms which vary with radial magnitude
- Decide on the generative backbone of different model components e.g., GANs or diffusion and flow-based approaches
- Train the full-distribution generative model
- Evaluate the resulting model on synthetic benchmarks

Takeaways

1. **Radial excesses above a high enough threshold and angular dependence structure become asymptotically independent,**
   Given the assumption of regular variation.

2. **The threshold is a difficult core aspect of extremal modelling.**
   Choosing u too low risks violating modelling assumptions, too high discards valuable tail observations.

3. **We identify two main class of approach.**
   Use a testing procedure to select u from the data (explored today) or develop flexible generative methods which avoid relying on a single threshold altogether (next steps).

Questions?

MGPD construction

$S \sim$ positive r.v., $\quad E_i \overset{\text{iid}}{\sim} \text{Exp}(1)$

$$\downarrow$$

$$W_i = \left(\frac{S}{E_i}\right)^{\theta} \qquad \text{Fréchet margins}$$

$$\downarrow$$

$$U_i = \exp\left(-\frac{1}{W_i}\right) \qquad U_i \sim \text{Unif}(0,1)$$

$$\downarrow$$

$$Z_i = \frac{\sigma_i}{\xi_i}\left[(1-U_i)^{-\xi_i} - 1\right] \qquad \text{GPD margins}$$

construction from Stephenson (2003)

